

Project Design Phase

Proposed Solution Template

Date: 15 February 2025

Team ID: LTVIP2025TMID35065

Project Name: Trafficelligence: Advanced Traffic Volume Estimation with Machine Learning

Maximum Marks: 2 Marks

*Proposed Solution Template:
Project team shall fill the following information in the proposed solution template.*

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Unpredictable and poorly managed urban traffic causes delays, congestion, and inefficiencies due to the lack of real-time forecasting models and adaptive systems.
2	Idea / Solution description	Trafficelligence leverages machine learning regression models to predict traffic volume by analyzing historical data, weather patterns, and time-based variables. The system enables better traffic management decisions by forecasting congestion in advance.
3	Novelty / Uniqueness	The solution integrates multiple ML models for comparative evaluation and optimizes traffic forecasts. It combines real-time data feeds and predictive analytics to improve traffic flow adaptability, which is not common in current static systems.

4	Social Impact / Customer Satisfaction	Reduces commuter stress and travel time, improves emergency vehicle mobility, supports public transportation scheduling, and helps reduce fuel consumption and emissions.
5	Business Model (Revenue Model)	Revenue can be generated by offering the system as a SaaS to city municipalities and transport departments. Additional licensing and support services can create continuous revenue.
6	Scalability of the Solution	The solution is scalable to any city or urban area by retraining the model with local traffic data. It can also be integrated with existing smart city infrastructure and IoT systems.